Song Energy Classifier

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Abstract

The classification of songs based on energy levels presents a multifaceted challenge within the domain of music information retrieval. This task requires analyzing audio features specifically tempo, loudness, speechiness, valence, and acousticness to determine the energy content of a song, a characteristic that significantly influences listener experience and engagement. Our research employs a comprehensive dataset of more than 30,000 tracks, leveraging advanced machine learning techniques including logistic regression, naive Bayes, support vector machines, random forest tree and k-Nearest Neighbors, tested against a training and validation split of 80:20. The goal of this research is to improve the accuracy of automated music recommendation systems by enabling more nuanced and responsive playlist curation based on dynamic listener preferences. Our findings demonstrate the potential of sophisticated modeling techniques to overcome the challenges of feature interaction and data imbalance, providing a pathway toward more personalized and engaging music streaming services. Through rigorous testing and validation, the project not only advances the understanding of audio feature analysis but also enhances the practical capabilities of music information retrieval systems to meet diverse user needs.

1. Introduction

The project focuses on the development of machine learning models to classify songs by their energy levels, which is crucial for enhancing user experience on music streaming platforms like Spotify and Apple Music. This classification helps tailor playlists for various activities and moods, benefiting not only individual users but also DJs, radio stations, and venues by aligning music selections with audience expectations. The task is challenging due to the subjective nature of what defines 'energetic' music, variability among listeners, and technical issues like data imbalance and recording quality differences. The goal is to accurately categorize the energy levels of over 30,000 songs, aiming to improve music recommendation systems and provide insights that could significantly enhance digital music interaction.

2. Related Work

Recent advancements in music information retrieval (MIR) have largely benefited from the integration of machine learning techniques, analyzing extensive datasets like those provided by Spotify. One notable study in this area is by Roberto Colangelo, titled "Machine Learning and Music: Predicting the level of energy conveyed by a soundtrack," which focuses on utilizing various machine learning algorithms to predict the energy levels in music tracks.

Colangelo's approach involves an in-depth classification of songs based on genre, employing a variety of linear, tree-based, nonlinear, and neural network models to analyze and predict music features. His methods include:

Linear Models: Simple Linear Regression, Ridge, and Lasso Regression.

Tree-Based and Nonlinear Models: Decision Trees, Random Forest, Gradient Boosting, and XGBoost.

Neural Network-Based Models: Multilayer Perceptron (MLP) and Deep Neural Networks (DNN).

Colangelo's results demonstrate significant achievements with tree-based and ensemble methods. For instance, the Random Forest Regressor, which builds multiple decision trees trained on random subsets of data, achieved an R-squared value of 0.863 and a Mean Absolute Error (MAE) of 0.071. Meanwhile, the XGBoost Regressor, known for its efficient handling of large datasets and incorporating techniques like regularization to mitigate overfitting, stood out with an R-squared of 0.878 and an MAE of 0.067, making it the most effective model in his study.

2.1. Illustrative Insights from Colangelo's Analysis:

Genre Classification Chart: A bar chart detailing the composition of genres within the top 100 most popular songs, indicating a prevalence of pop and hip-hop, which are critical to understanding genre-specific energy dynamics.

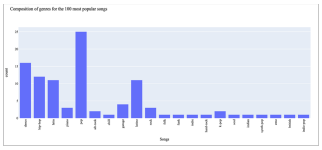

Figure 1. Composition of genres for the 100 most popular songs

Danceability by Genre: A boxplot showing the distribution of danceability across genres, highlighting how different musical styles influence listener perception and physical response.

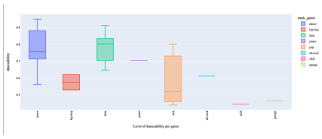

Figure 2. level of danceability per genre

Energy vs. Loudness: A scatter plot with a linear regression fit illustrating the relationship between loudness and the energy level of tracks, which is pivotal for understanding how these audio features correlate.

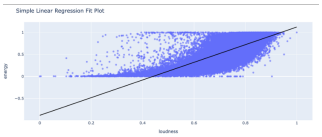

Figure 3. Simple linear regression fit plot

Comparison with Our Methodology: In contrast to Colangelo's broad application of multiple complex models, our study narrows down to a selected few features that significantly impact a song's energy. We focus on traditional methods like Linear Regression, Random Forest, SVM, Naïve Bayes, and KNN to classify songs by energy levels, optimizing for both accuracy and computational efficiency.

This comprehensive analysis and the methodologies applied in both studies underscore the potential of machine learning in transforming the music industry by enabling more nuanced and predictive analytics of music characteristics.

2.2. Margins and page numbering

3. Method

The methodology for classifying songs based on their energy levels is a multifaceted approach that integrates data preprocessing, feature analysis, machine learning model selection, training, and validation. Here is a detailed walkthrough of the methodology employed in this study.

3.1. Data Collection and Preparation

Dataset Acquisition: The dataset comprises over 30,000 songs with various audio features and labels indicating energy levels. This data was sourced to ensure a diverse representation of genres, artists, and production styles.

Feature Selection: Key audio features considered for the study include loudness, tempo, loudness, speechiness, valence, and acousticness. These features are known to influence the perceived energy of music and are thus critical for the classification task.

Data Cleaning: The dataset was cleaned to remove any missing or corrupt data entries. This step ensures the robustness and reliability of the training process and subsequent analysis.

3.2. Feature Analysis and Preprocessing

Statistical Analysis: Initial statistical analysis was performed to understand the distribution, mean, median, and standard deviation of the features. This step helps identify outliers and understand feature scales.

Normalization: To address the scale disparity among features, Min-Max scaling was applied to transform all feature values to a uniform scale between 0 and 1. This normalization facilitates better performance in machine learning algorithms, especially those sensitive to feature scaling like.

3.3. Model Selection

A variety of machine learning models were employed to evaluate their performance in predicting the energy levels of songs

K-Nearest Neighbors (KNN): Chosen for its effectiveness in classification tasks where the class label of a data point is determined by the majority label among its nearest neighbors. The model was fine-tuned using GridSearchCV to identify the optimal number of neighbors.

Linear Regression: Employed as a baseline to ascertain the linear relationship between features and the song's energy level.

Random Forest Tree: A robust model that constructs multiple decision trees during training and outputs the class that is the mode of the classes of the individual trees, offering benefits in handling overfitting and providing feature importance insights**.**

Support Vector Machine (SVM): Utilized for its capacity to find the hyperplane that best divides a dataset into classes in the feature space.

Naive Bayes: Applied due to its assumption of independence between predictors, Naive Bayes is a simple yet effective probabilistic classifier.

3.4. Model Training

The models were trained using a split of the data into training and testing sets, typically with an 80-20 ratio. The training set was used to train the models, while the testing set helped evaluate their generalizability to new data.

3.5. Model Evaluation

Each model's performance was assessed using several metrics:

Accuracy: Measures the overall correctness of the model.

Precision, Recall, and F1 Score: These metrics provide a more detailed insight into the model's performance, especially in scenarios where class imbalance might be present.

Confusion Matrix: Offers a visualization of the performance of the model in terms of the number of correct and incorrect predictions divided by class.

4. Experiments

4.1. Dataset Description

The data set used for this study consists of music tracks, where each track is described by multiple attributes but we have chosen attributes such as loudness, speechiness, acousticness, valence, and tempo. These features are commonly associated with the perceived energy of music, which the model attempts to classify into "High Energy" or "Low Energy" categories.

4.2. Experiment Setup Of Logistic Regression

The logistic regression model was selected for its efficiency in binary classification tasks. The model was configured with the following settings:

Solver: Liblinear — appropriate for small to medium datasets.

Regularization: L2 — to help prevent the model from overfitting.

C (Regularization strength): 1.0 — standard value to start with regularization.

The training process involved using a subset of the data to train the model, followed by validation using a separate test set.

4.2.1 Evaluation Metrics

Confusion Matrix: The confusion matrix shows the number of correct and incorrect predictions made by the model. The matrix indicates

True Positives: 1480
True Negatives: 1141
False Positives: 552
False Negatives: 319

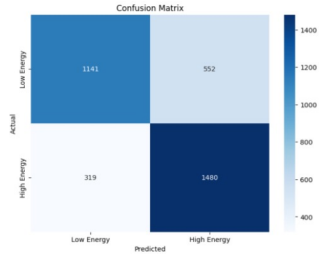

Figure 4. Confusion Matrix of Logistic Regression

```
Accuracy: 0.7505727376861397

Confusion Matrix:
[[1141  552]
 [ 319 1480]]

Classification Report:
              precision    recall  f1-score   support

       False       0.78      0.67      0.72      1693
        True       0.73      0.82      0.77      1799

    accuracy                           0.75      3492
   macro avg       0.75      0.75      0.75      3492
weighted avg       0.75      0.75      0.75      3492
```

Figure 5. Classification Metrics

4.2.2 Result

The model achieved an accuracy of approximately 75.05%, with the weighted averages of precision, recall, and F1-score all aligning closely**.** From feature importance we can see Notably, 'loudness' is the most influential feature, significantly impacting the model's predictions, suggesting that louder tracks are generally perceived as higher energy.

4.3. Experiment Setup Of Random Forest Tree

Model Training: A Random Forest classifier was employed with the hyper-parameters set to their default values except for the random state, which was fixed at 42 to ensure reproducibility.

4.3.1 Evaluation Metrics

Confusion Matrix:

True Negative (Low Energy, Predicted Low): 2056
False Positive (Low Energy, Predicted High): 910
False Negative (High Energy, Predicted Low): 578
True Positive (High Energy, Predicted High): 3022

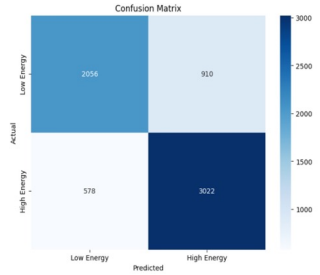

Figure 6. Confusion Matrix of Random Forest Tree

Model Performance: The Random Forest classifier achieved an accuracy of approximately 77.37%. The precision scores were 0.78 for classifying low energy songs

```
Accuracy: 0.7733780079195858

Confusion Matrix:
[[2056  910]
 [ 578 3022]]

Classification Report:
              precision    recall  f1-score   support

       False       0.78      0.69      0.73      2966
        True       0.77      0.84      0.80      3600

    accuracy                           0.77      6566
   macro avg       0.77      0.77      0.77      6566
weighted avg       0.77      0.77      0.77      6566
```

Figure 7. Classification Metrics

and 0.77 for high energy songs. Recall scores were 0.69 for low energy and 0.84 for high energy, leading to F1-scores of 0.73 and 0.80 respectively.

4.3.2 Result

The model achieved an accuracy of approximately 77.37%, indicating a reasonably high level of performance in classifying songs based on their energy levels. The most influential features in classification were found to be 'loudness' and 'acousticness', after feature doing feature importance.

4.4. Experiment Setup Of Support Vector Machine

In data preprocessing, apart from the predetermined steps, the 'energy' feature was binarized based on a threshold of 0.7, where values above this threshold are considered high-energy. In the model configuration, the Support Vector Machine (SVM) model was employed with a default linear kernel and Hyper-parameters were set to their default values, with the model trained on the training set.

4.4.1 Evaluation Metrics

Confusion Matrix:

True Negatives (TN): 3171
False Positives (FP): 1306
False Negatives (FN): 756
True Positives (TP): 4617

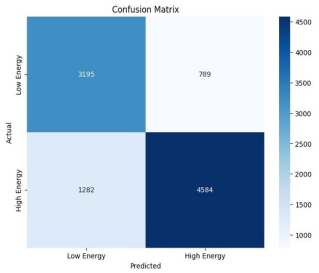

Figure 8. Confusion Matrix of Support Vector Machine

4.4.2 Result

The SVM model achieved an accuracy of approximately 79.07%. This metric indicates the overall effectiveness of the model in classifying the songs correctly as high or low energy. From doing feature importance, we can see that

```
SVM Confusion Matrix:
[[3171 1306]
 [ 756 4617]]
SVM Accuracy: 0.7906598984771573
SVM Precision: 0.7795036299172716
SVM Recall: 0.8592964824120602
SVM F1 Score: 0.8174575070821529
```

Figure 9. Classification Metrics

Loudness was identified as the most influential feature, followed by 'acousticness' and 'valence'. This suggests that louder songs with less acousticness tend to have higher energy.

4.5. Experiment Setup Of Naive Bayes

In data preprocessing, apart from the predetermined steps, the 'energy' attribute was converted to a binary format, where values greater than a 0.7 threshold were labeled as high energy. In model configuration, a Gaussian Naive Bayes classifier was utilized and no specific hyperparameters were adjusted for the Naive Bayes model.

4.5.1 Evaluation Metrics

Confusion Matrix:

True Negatives (TN): 2660
False Positives (FP): 1817
False Negatives (FN): 708
True Positives (TP): 4665

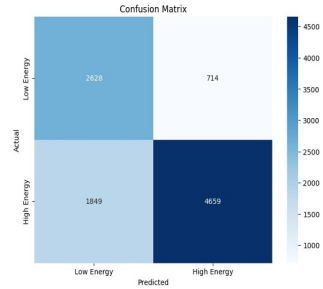

Figure 10. Confusion Matrix of Naive Bayes

```
Naive Bayes Confusion Matrix:
[[2660 1817]
 [ 708 4665]]
Naive Bayes Accuracy: 0.7436548223350253
Naive Bayes Precision: 0.7196852823202715
Naive Bayes Recall: 0.8682300390843104
Naive Bayes F1 Score: 0.7870097005482919
```

Figure 11. Classification Metrics

4.5.2 Result

The Gaussian Naive Bayes model achieved an accuracy of approximately 74.36%. And feature importance indicates 'Acousticness' and 'loudness' emerged as the most significant features affecting the prediction, indicating that these

features play a crucial role in distinguishing between high and low energy songs.

4.6. Experiment Setup of K-nearest neighbor

The model's hyperparameters were optimized using grid search, specifically focusing on the number of neighbors (k). The range of k was set from 1 to 10, and the model's performance was evaluated through 5-fold cross-validation to ensure robustness against overfitting. The primary hyperparameter tuned was 'n_neighbors', the count of nearest neighbors considered for voting in the classification.

4.6.1 Evaluation Metrics

Confusion Matrix: True Positives (TP): 3802 True Negatives (TN): 1524 False Positives (FP): 773 False Negatives (FN): 468

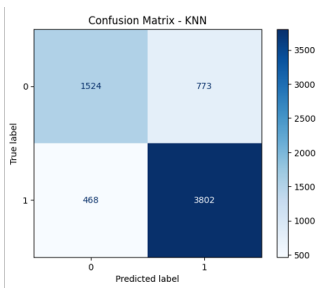

Figure 12. Confusion Matrix of K-nearest neighbor

```
Best k: 9
[[1524  773]
 [ 468 3802]]
Accuracy: 0.8110248210750723
Precision: 0.8310382513661202
Recall: 0.8903981264637002
F1 Score: 0.8596947427925381
```

Figure 13. Classification Metrics

4.6.2 Result

These results yielded an accuracy of 81.10%, precision of 83.10%, recall of 89.39%, and an F1 score of 85.96% Indicating this is the best model as compared to others The feature importance chart, based on the mean decrease in accuracy when excluding each feature, highlighted that 'loudness' and 'acousticness' were the most significant predictors in classifying song energy, suggesting these attributes have a substantial impact on the perceived energy of music

5. Conclusion

From our analysis we can clearly see that KNN performed much better as it focuses on the similarity between closest data points, making it well-suited for datasets where instances with similar features (e.g., acousticness, loudness) share the same label (e.g., energetic or non-energetic). Features like loudness and tempo are likely to have clear separations between energetic and non-energetic songs, enabling KNN to classify effectively. Normalization is crucial for KNN thus Min-Max scaling ensured that all features contributed equally to the distance metric, preventing large-scale features (e.g., tempo) from dominating small-scale ones (e.g., speechiness). It uses proximity-based voting, making it effective for datasets with clear relationships between features and target labels. In summary, our project bridges the gap between the technical attributes of music and the emotional connection listeners experience. By classifying songs based on their 'vibes,' we've opened up new possibilities for personalized music discovery. Looking ahead, the integration of user feedback and real-time mood detection could make this system even more adaptive and personalized

References

· https://youtu.be/0p0o5cmgLdE?si=pMOg4NgyH9mNcKno for K-nearest neighbor

· kaggle for data set.

· Research paper on Machine Learning and Music: Predicting the level of energy conveyed by a soundtrack by Roberto Colangelo for related works :